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Numerical simulations of a parabolic trough solar collector with nanofluid using a two-phase model

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ABSTRACT

This study investigates numerically, in terms of Computational Fluid Dynamics (CFD), a Parabolic Trough Collector (PTC) system with nanofluid as the Heat Transfer Fluid (HTF). All the heat transfer mechanisms were taken into account to simulate the SEGS LS2-module PTC. The validation process showed very good agreement between the numerical results and the available test results from four typical testing conditions, with the use of Syltherm 800 liquid oil. Specifically, the maximum relative error observed for outlet temperature was 0.3% and 7.3% for the collector efficiency. In order to address the nanofluid modeling problem the two-phase approach was preferred (against single-phase) and validated against experimental and numerical results for a circular tube under constant wall temperature. Overall, a total of 20 different simulation cases were performed for the LS2 module, for a range of nanoparticle (Al_2O_3) concentrations (0%–4%), thus making it possible for a parametric evaluation on the LS2 efficiency. In addition, the temperature and velocity fields of the Syltherm 800/Al_2O_3 nanofluid were associated with the enhanced heat transfer occurring at higher nanoparticle concentrations. A boost up to 10% on the collector efficiency was reported for Al_2O_3 concentration of 4%, which is in accordance with relevant studies.

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1. Introduction

1.1. Parabolic trough solar collectors – numerical modeling

Parabolic trough solar collectors (PTC) are systems with light structures and low cost technology that deliver high temperatures with good efficiency. A metal tube, covered with a glass tube to reduce heat losses, is placed along the focal line of the receiver (Fig. 1). Several prototype and commercial PTCs were manufactured over the last four decades and relevant experimental studies were conducted. One of the types most extensively investigated, being also the PTC type of interest here, is the SEGS LS-2 module, for which experimental data were obtained by Dudley et al. [1], aiming at calculating its thermal performance and heat losses.

Furthermore, over the past decade, ever increasing capabilities have become available for also addressing PTC systems numerically, in terms of Computational Fluid Dynamics (CFD). The relevant scientific studies focused mainly on two different flow configurations and respective computational domains: one that comprises the absorber tube and the heat transfer fluid, and a second type that addresses the entire PTC system (absorber tube, heat transfer fluid, glass cover and annulus space between absorber and glass cover).

Concerning the first type, He et al. [2] and Sadaghiyani et al. [3] used a coupled simulation method based on Monte Carlo Ray Tracing (MCRT) and CFD software to solve the complex coupled heat transfer problem in the LS-2 PTC, evaluating among others the collector performance. Studies that belong in the second category include, Cheng et al. [4] who used numerical simulations of coupled heat transfer characteristics in the receiver tube in combination with the MCRT method and the FLUENT software, in which the heat transfer fluid and physical model were Syltherm 800 liquid oil and LS2 parabolic solar collector from the testing experiment of Dudley et al. [1], respectively. Cheng et al. [5] extended the work of Cheng et al. [4] studying typical heat transfer fluid types and residual gas conditions.

As in the aforementioned studies, Islam et al. [6] also used LS2 [1] module to perform conjugate heat transfer simulations for a concentrating solar thermal collector, with modeling heat transfer performances of faceted absorbers for concentrating photovoltaic

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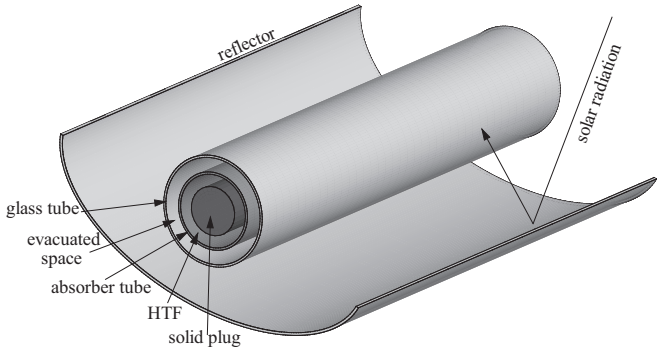


Fig. 1. Physical model of the LS-2 parabolic trough solar collector.

applications in mind. Mwesigye et al. [7] conducted numerical simulations for the LS2 module in which they added perforated plate inserts. Wang et al. [8] used a solar ray trace method and the finite element method (FEM) to solve the complex problem coupled with fluid flow, heat transfer and thermal stress in a PTC system.

1.2. Enhanced thermal behavior of solar systems using nanofluids

Recent studies have shown that a promising technique to improve thermal conductivities of heat transfer fluids and therefore to enhance heat transfer, is to suspend solid particles of nanoscale dimensions in the fluids to create suspensions known as *nanofluids*. Even though the anomalous increase in the thermal conductivity with respect to the base fluid presented in early studies on nanofluids could not be confirmed in later experiments (such as the benchmark study of Buongiorno [9] with over 30 participants worldwide), research on the application of nanofluids in various fields, among them solar energy utilization, has been steadily growing over the past few years. This is reflected in quite a few recent literature reviews [10–14]. Aside from the original focus on thermal conductivity, other features of nanofluids such as optical properties, efficiency and transmission as well as extinction coefficients of solar systems have made them more attractive in this field compared to conventional fluids [13]. Some limitations are certainly also present, such as instability, agglomeration, increased pumping power, erosion and corrosion of the heat transfer equipment [14].

1.2.1. Experimental studies

Flat plate solar collectors occupy a large portion of the relevant literature on experimental studies carried out to this day, whereby efficiencies have been found to increase by up to 30% with the use of nanofluids [13]. The literature also shows that a much investigated system of application is the direct absorption collector as well as the photovoltaic-thermal (PVT) collector, where the enhancement is more significant [14]. On the other hand, limited number of studies have been conducted on application of nanofluids in concentrating collectors such as the PTC.

Jafar and Sivaraman [15] carried out an experimental investigation of a PTC with a Al_2O_3/water nanofluid in conjunction with nail twisted tape inserts, under laminar flow conditions. Heat transfer was found to increase through the use of the nanofluid, with a further enhancement due to the effect of the twisted tapes. More recently Kasaeian et al. [16] in an experimental study have considered a PTC with multi walled carbon nanotube (MWCNT)/oil based nanofluid as the working fluid. Their results showed an efficiency improved by up to 7% compared to pure oil. Turkyilmazoglu [17] also studied the heat transfer enhancement effects of nanofluids in circular concentric pipes under the influence of partial velocity slips on the surfaces. He used a single phase nanofluid model to get the fully developed exact laminar flow and temperature fields.

1.2.2. Computational studies

Concerning now computational studies that include the use of nanofluids with configurations that are close to the LS2 module, Moghari et al. [18] studied heat transfer enhancement of a mixed convection, laminar Al_2O_3/water nanofluid flow, in an annulus with constant heat flux boundary condition employing two phase mixture model. Their simulation results showed that at a given Reynolds (*Re*) and Grashof (*Gr*) numbers, increasing nanoparticles volume fraction increases the Nusselt number and that the inner wall had larger values than their corresponding values at the outer wall.

Kasaeian et al. [19] studied the fully developed turbulent mixed convection heat transfer of Al_2O_3/synthetic oil nanofluid numerically (with single-phase approach) in a PTC with uniform heat flux. Their results showed that the Nusselt numbers and the convection heat transfer coefficients have a direct dependency on the volume fraction of the nanofluid.

Khullar et al. [20] have considered also the enhancements arising from the improved optical properties of a nanofluid (absorption, scattering) in their modeling of a parabolic collector with aluminum nanoparticles dispersed in Therminol-VP1. They developed a two-dimensional heat transfer model which incorporated volumetric absorption effects in terms of absorption and scattering mechanisms. Improvements in efficiency of 5–10% were found compared to a conventional concentrating parabolic collector.

Sokhansefat et al. [21] studied the three-dimensional fully developed turbulent mixed convection heat transfer of Al_2O_3/synthetic oil nanofluid in a trough collector tube with a non-uniform heat flux. They also investigated the effect of Al_2O_3 particle concentration in the synthetic oil on the rate of heat transfer from the absorber tube, using the FLUENT software and the single-phase approach in order to simulate the effect of the nanoparticles.

It becomes evident from the above that there is a very limited number of studies incorporating nanofluids in PTCs. Furthermore, the use of the two-phase approach in modeling nanofluids, has been shown in comprehensive relevant review studies, for instance the one by Vanaki et al. [22], to give more accurate results in most problems investigated compared to the single-phase approach. More specifically, in Ref. [22], comparisons between the two approaches were reviewed for single- and two-phase nanofluids, for laminar and turbulent forced and mixed convection in various geometries against experimental data. The performance of the two models was quite problem-dependent and determined by the Reynolds number, the flow regime and the particle concentration in particular. Generally, however, where quantitative data for the discrepancies were provided, values for the two-phase model were mostly less than 10%, whereas for the single-phase, homogeneous model the respective values, even though small in some cases, were mostly higher than 10% with values as high as 30% reported in one study.

In view of the above, the goal of the present study is to approach numerically, the use of nanofluids in a common PTC module (such as the LS-2 of Dudley et al. [1]), with the more accurate two-phase approach and examine the fluid flow and heat transfer characteristics of such configuration. Furthermore, realistic thermal boundary conditions are employed, taking into account the actual radiative flux distribution on the tube absorbing surface using an optical analysis model.

2. PTC model and numerical approach

2.1. Parabolic trough solar collector

The LS-2 PTC module is chosen as the model for the present simulations, since it has formed the basis for many of the PTCs that were designed afterwards [23]. Besides, a large amount of relevant experimental data are available for validation from the study of Dudley et al. [1]. The schematic diagram of the physical model is shown in Fig. 1. The receiver is composed of an outer glass tube, an inner absorber tube with cermet selective coating on the surface and a solid plug restricting device, inserted at the center of the receiver.

The physical mechanisms of heat transfer evolved in the particular configuration are as follows [5]:

- conduction heat transfer in the solid plug
- convection heat transfer between the solid plug and the HTF
- convection heat transfer between the HTF and the absorber
- solar irradiation absorption in the absorber
- conduction heat transfer through the absorber wall
- radiation from the absorber to the glass tube
- solar irradiation absorption in the glass tube
- conduction heat transfer through the glass tube
- heat transfer from the glass tube to the atmosphere

The length of the receiver tube is $L = 7.8$ m and the mirror aperture width is 5 m resulting 39 m^2 aperture window. As seen in Fig. 2, the detailed set of radius data is: $r_1 = 0.0254$ m, $r_2 = 0.0330$ m, $r_3 = 0.0350$ m, $r_4 = 0.0545$ m, $r_5 = 0.0575$ m.

2.2. Materials

In the present section the properties of each material of the PTC needed for the numerical simulations are described in detail.

Glass tube. The glass tube is made from Pyrex glass anti-reflective coated Dudley et al. [1]. The properties of Pyrex glass Corning [24], Sri Prasad Labglass Technologies Ltd. [25]. can be found in Table Appendix A.1.

Solid plug. The material for the solid plug is stainless steel and its thermal conductivity is fixed at 54 $Wm^{-1}\ K^{-1}$ [2].

Absorber. The absorber is made from copper and has a cermet selective surface of 0.96 absorptance and 0.14 emittance at 350 *Co* Dudley et al. [1].

Vacuum on receiver annulus. According to the experiments conducted by Dudley et al. [1], three different configurations were tested concerning the annulus between the receiver and the glass envelope: vacuum, air and glass envelope completely removed. In the present study the scenario with vacuum is considered.

Heat transfer fluid: *Syltherm 800*. In the experimental work of Dudley et al. [1] Syltherm 800 oil was used as the heat transfer fluid [26]. The properties of Syltherm 800 could be approximated by a polynomial of the form:

$$\text{Property} = a_0 + a_1 T[K] + a_2 T^2[K] + a_3 T^3[K] + a_4 T^4[K] \quad (1)$$

Coefficients of Eq. (1) for each parameter were extracted from Dow Chemical Company [26] data and can be found in Table Appendix A.2. They are valid for the temperature range 300 – 650 [K]. The parameters for dynamic viscosity were taken from Ref. [27].

Nanoparticles: Al_2O_3. The authors were able to find only two studies in the literature where nanofluids are used in PTCs as HTFs. The first one, by Kasaeian et al. [16], was experimental and the authors used carbon nanotubes as nanoparticles in a custom-made PTC. The second one, by Sokhansefat et al. [21], was a numerical study on a LS2 module PTC (same as the one under study here) were Al_2O_3 nanoparticles were used by the authors.

In the present study Al_2O_3 nanoparticles will also be used. Their properties may be found in Table Appendix A.3, which is in accordance to Sokhansefat et al. [21]. It should be noted that since here the emphasis was on the modeling techniques and on assessing the performance of the two-phase relative to the single-phase approach in this problem, the effect of the nanoparticle material was considered outside the scope of this study. Several review studies, however, have reported extensively on this effect, with regard to solar systems in particular, which the interested reader may refer to (e.g. Verma and Tiwari [14]).

The justification for selecting the value $\mu = 10^{-8}$ [Pa s] for the dynamic viscosity is provided in the following section.

Nanofluid: Al_2O_3/*Syltherm 800*. The nanofluid properties can be evaluated from the following equations: The correlation of effective density is taken from Pak and Cho [28]:

$$\rho_{nf} = (1-\phi)\rho_f + \phi\rho_s \quad (2)$$

The correlation for the effective viscosity is taken from Einstein [29]:

$$\mu_{nf} = (1+2.5\phi)\mu_f \quad (3)$$

The correlation for the effective conductivity is taken from Yu and Choi [30]:

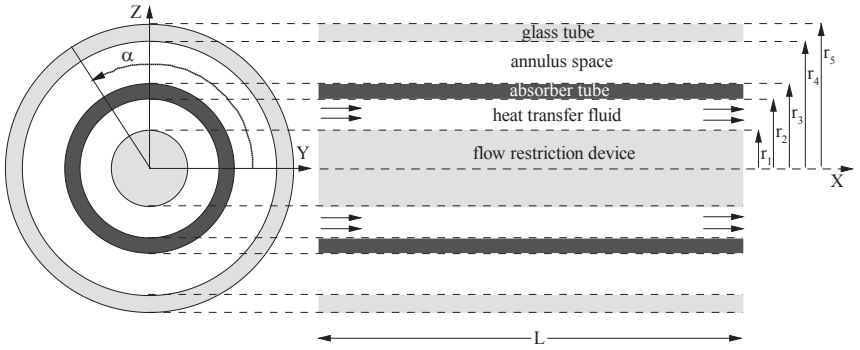


Fig. 2. Schematic of the LS2 parabolic trough solar collector computational model.

$$k_{nf} = k_f \frac{k_s + 2k_f - 2(1+0.1)^3\phi\left(k_f - k_s\right)}{k_s + 2k_f + (1+0.1)^3\phi\left(k_f - k_s\right)} \tag{4}$$

2.3. Numerical modeling

In order to simulate the nanofluid flow, two techniques were used in the present study. The first one, used only in the validation case (Section 3), is the single-phase approach, which assumes that base fluid and nanoparticles have the same temperature and velocity field. Therefore, continuity, momentum and energy equations can be solved as if the fluid were a classical Newtonian fluid by using effective properties for the nanofluid.

The second technique was based on the Eulerian-Eulerian single fluid, two-phase approach [31], assuming that the coupling between phases is strong, and particles closely follow the flow [32]. The two phases are assumed to be inter-penetrating, meaning that each phase has its own velocity vector field, and within any control volume there is a volume fraction of primary phase and also a volume fraction of the secondary phase. This model has been shown to give successful predictions even for low particle concentrations [33]. Instead of utilizing the governing equations of each separately, the continuity, momentum and energy equations for the mixture are employed [34]. The continuity equation is given as (in the following equations $\vec{U}$ is represented as U):

$$\nabla(\rho_m U_m) = 0 \tag{5}$$

where, the mass-averaged velocity or mixture velocity, U_m, is defined as:

$$U_m = \frac{\rho_s \phi_s U_s + \rho_f \phi_f U_f}{\rho_m} \tag{6}$$

where U_s is the particle velocity, U_f is the base fluid velocity and ρ_m is the mixture density for two-phase mixture that is defined as:

$$\rho_m = \rho_s \phi_s + \rho_f \phi_f \tag{7}$$

The steady state momentum equation is:

$$\rho_m(U_m \nabla U_m) = -\nabla P + \mu_m\left(\nabla U_m + (\nabla U_m)^T\right) + \nabla\left[\phi_f \rho_f U_{dr,f} U_{dr,f} + \phi_s \rho_s U_{dr,s} U_{dr,s}\right] + \rho_m g \tag{8}$$

where, P is the pressure, μ_m is the viscosity of mixture, $U_{dr,f}$ and $U_{dr,s}$ are the drift velocity of particles and base fluid respectively:

$$U_{dr,p} = U_p - U_m \quad \text{and} \quad U_{dr,f} = U_f - U_m \tag{9}$$

The steady state energy equation is given as:

$$\nabla\left(\phi_s U_s \rho_s h_s + \phi_f U_f \rho_f h_f\right) = \nabla\left[\left(\phi_s k_s + \phi_f k_f\right)\nabla T\right] \tag{10}$$

where, h_f and h_s are the enthalpy of base fluid and solid particles respectively. The volume fraction equation for two-phase mixture is:

$$\nabla(\phi_s \rho_s U_m) = -\nabla(\phi_s \rho_s U_{dr,s}) \tag{11}$$

The slip velocity is defined as:

$$U_{f,s} = U_f - U_{f,s} \tag{12}$$

and the relation between drift velocity and relative velocity can presented as:

$$U_{dr,s} = U_{s,f} - \frac{\rho_s \phi_s}{\rho_m} U_{f,s} \tag{13}$$

The relative velocity is through the Schiller and Naumann [35] drag formulation:

$$U_{f,s} = \frac{d_p^2}{18\mu_f f_d} \frac{(\rho_s - \rho_m)}{\rho_s} \vec{\alpha} \tag{14}$$

$$f_d = 1 + 0.15 Re_s^{0.687} \quad \text{for } Re_s \le 1000 \quad \text{and } f_d = 0.0183 Re_s \text{ otherwise} \tag{15}$$

$$\vec{\alpha} = \vec{g} - (U_m \nabla) U_m \tag{16}$$

where, $\vec{\alpha}$ and $\vec{g}$ are the particle's and fluid gravitational acceleration respectively. The particle Reynolds number (Re_s) is defined as:

$$Re_s = \frac{U_m d_p \rho_m}{\mu_m} \tag{17}$$

where d_p is the mean particle diameter, here 38 [nm].

In all simulation cases of the present study, the flow of the HTF inside the receiver is turbulent since the Re is greater than 2300 (details in Section 2.5). In order to simulate accurately the resulting turbulent flows inside the annulus, apart from the continuity, momentum and energy equations, the equations for the k-ε model were also employed in the ANSYS − CFX solver ANSYS Inc. [31]. The choice of the k-ε turbulence model was based on its widely acceptance, since this has been successfully used in most of the relevant numerical studies PTCs [2–5,19,21] The temperature-depended thermo-physical properties of the HTF were also taken into account in the simulations. The equations that describe the k-ε model are:

$$\nabla(\rho_m U_m k) = \nabla\left[\left(\mu_m + \frac{\mu_{t,m}}{\sigma_k}\right)\nabla k\right] + G_{k,m} - \rho_m \varepsilon \tag{18}$$

$$\nabla(\rho_m U_m \varepsilon) = \nabla\left[\left(\mu_m + \frac{\mu_{t,m}}{\sigma_\varepsilon}\right)\nabla \varepsilon\right] + \frac{\varepsilon}{k}(c_1 G_{k,m} - c_2 \rho_m \varepsilon) \tag{19}$$

where the turbulent viscosity μ_t and the production rate of k, G are given by

$$\mu_{t,m} = C_\mu \rho_m \frac{k^2}{\varepsilon}, \quad G_{k,m} = \mu_{t,m}\left(\nabla U_m + (\nabla U_m)^T\right) \tag{20}$$

The standard constants are employed, $C_\mu = 0.09$, $c_1 = 1.44$, $c_2 = 1.92$, $\sigma_k = 1.0$, $\sigma_\varepsilon = 1.3$ and $\sigma_t = 0.85$.

Radiation modeling in the annulus space was treated by the Monte Carlo method [31], whereby the radiation is assumed to affect the medium by heating or cooling the surface of the domain, with no radiant energy transfer directly to the medium (Surface-to-surface transfer mode). This assumption is valid since (as already mentioned in Section 2.2) the annulus space was considered as filled with air at very low pressure (1 torr). The spectral dependency of the radiative transfer equation was approximated with the Gray model which assumes that all radiation quantities are nearly uniform throughout the spectrum.

The steady-state version of the coupled governing equations were used and higher-order spatial discretization schemes were considered. The convergence criterion for the solver variables of the flow and energy is the RMS residual to be less than 10^{-6}.

2.4. Boundary conditions

The boundary conditions used for the numerical simulations of LS-2 PTC were:

Solid plug. Adiabatic boundary conditions on inlet and outlet surfaces.

HTF domain. Uniform inlet velocity and temperature. Fully-developed flow assumption on outlet ($\partial u/\partial x = \partial v/\partial x = \partial w/\partial x = \partial T/\partial x = 0$). Velocity is set to zero on both inner and outer side surface of the HTF domain.

Absorber. Adiabatic boundary conditions on inlet and outlet surfaces. Non-uniform solar heat flux on the outer side surface (see Section 2.4.1).

Annulus space. Adiabatic boundary conditions on inlet and outlet surfaces. Radiation boundary conditions as described earlier with surface-to-surface model. Velocity of the nearly evacuated air space is set to zero on both inner and outer side surfaces.

Outer glass. Adiabatic boundary conditions on both inlet and outlet surfaces. A convection boundary condition was used on the outer side surface to model heat losses due to heat convection from the wind. Mullick and Nanda [36] proposed an equation for predicting the overall heat loss factor, later also used with success by Naeeni and Yaghoubi [37] in their numerical simulations. The equation is:

$$h_{wind} = 4V_{wind}^{0.58} D_{g,o}^{-0.42} \tag{21}$$

where V_{wind} is the wind velocity in [m s^{-1}] and $D_{g,o}$ is the outer glass diameter in [m].

2.4.1. Solar heat flux – local concentration ratio

Cheng et al. [5] calculated the local concentration ratio (LCR) on the outer surface of the absorber with a Monte Carlo Ray-Trace method (Fig. 3). In the present study a 7-curve fit was used to approximate the local concentration ratio also shown in Fig. 3. The curves are 3rd degree polynomial functions of the form:

$$\text{LCR} = b_3\phi^3 + b_2\phi^2 + b_1\phi + b_0 \tag{22}$$

and the coefficients for each segment can be found in Table 1.

2.5. Test cases

Four test cases from the Dudley et al. [1] experimental data were chosen to validate the code at first and then nanoparticles were added to the HTF, over a wide range of operating conditions ($T_{in} = 375 - 570$ [K]). The test cases correspond to cermet selective coating on the receiver surface and vacuum in the annulus region. In Table 3 the resulting dimensionless numbers (*Re*, *Gr* [38], *Pr*, *Ri*) for the arising mixed convection heat transfer problem are presented for each case.

Regarding the various regimes arising in mixed convection in horizontally oriented tubes, the most comprehensive study is the one by Metais and Eckert [39] who studied heat transfer in laminar to turbulent flows under uniform wall temperature. Fig. 4 shows the various regimes and their limits. The authors were unable to find a similar regime map for annulus under constant wall heat flux. Nevertheless, the use of the aforementioned map is only for qualitative purposes.

According to Turner [40], a rule of thumb to define the convection regime, is to calculate the Richardson number, *Ri*, which expresses the ratio of buoyancy to inertia forces ($Ri = Gr/Re^2$). For pure forced convection $Ri \ll 1$, for pure natural convection $Ri \gg 1$ and for mixed convection flows $Ri \simeq 1$.

Both approaches agree with the numerical results of [3] that the test cases under consideration belong to the turbulent flow regime and mixed convection effects (such as secondary flows, assisting heat transfer) are expected.

Studies that also include mixed convection flows in an annulus have been carried out, among others, by Habib and Negm [41] and Islam et al. [42] who investigated numerically the steady, laminar, mixed convection in the fully developed region of horizontal concentric annuli for the cases of non-uniform heating. They both found that non-uniform heating arrangements give rise to secondary flows.

2.6. Efficiency of the parabolic trough collector

For characterizing the performance of the nanofluid as heat transfer fluids, the efficiency of the parabolic trough collector is calculated, based on the equation:

$$\eta\% = \frac{\dot{m}_{in}\overline{C}_p(T_{out} - T_{in})}{A_{ap}I_b} \tag{23}$$

where $\dot{m}_{in} = \rho_{in}\dot{V}_{in}$ is the inlet mass flow rate in [kg s^{-1}], $\overline{C}_p = \frac{1}{2}(C_{p,in} + C_{p,out})$ is the specific heat capacity average in inlet and outlet, T_{in}, T_{out} are the mean temperatures in inlet and outlet

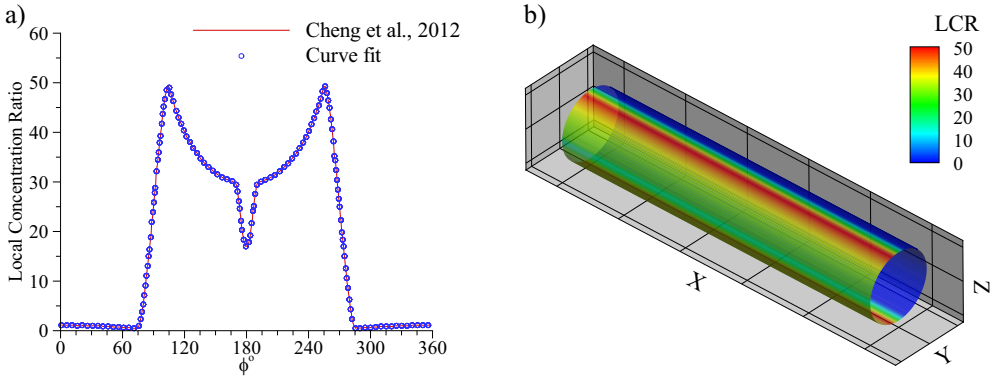


Fig. 3. The local concentration ratio on the outer surface of the absorber tube, as calculated by Cheng et al. [5] (a) and a 3-D representation of the local concentration ratio on the outer surface of the absorber tube (b).

Table 1
Coefficients of the LCR curve fitting.

Segment	a range	b_3	b_2	b_1	b_0
1	0–75	0.0	−1.071170E-4	−8.100954E-4	1.112046
2	75–104	−2.544403E-3	6.878607E-1	−5.974390E1	1.685403E3
3	104–171.2	−6.602394E-5	3.196692E-2	−5.280388	3.275329E2
4	171.2–188.8	0.0	1.524597E-1	−5.488588E1	4.957224E3
5	188.8–256	5.961826E-5	−3.504845E-2	6.979938	−4.403785E2
6	256–285	2.493475E-3	−2.019052	5.427366E2	−4.840387E4
7	285–360	0.0	−7.511141E-5	5.688045E-2	−9.606886

Table 2
Test cases. Experimental data from Dudley et al. [1].

Case	I_b [W m^{-2}]	U_{wind} [m s^{-1}]	T_{air} [K]	T_{in} [K]	T_{out} [K]	ΔT_{air} [K]	$q_{m,in}$ [kg s^{-1}]	η %	Est. error ±%
1	968.2	3.7	295.55	424.15	446.45	139.8	0.6522	70.90	1.92
2	982.3	2.5	297.45	470.65	492.65	184.3	0.6348	70.17	1.81
3	909.5	3.3	299.35	523.85	542.55	233.9	0.6596	70.25	1.90
4	937.9	1.0	301.95	570.95	590.05	278.6	0.6229	67.98	1.86

Table 3
The resulting dimensionless number from the experimental data of Dudley et al. [1].

Case	I_b	Re	Gr	Pr	Ri
1	968.2	4761	3.1×10^7	25.6	1.37
2	982.3	7196	7.8×10^7	18.7	1.51
3	909.5	11,207	1.7×10^8	14.4	1.34
4	937.9	15,044	3.6×10^8	11.7	1.60

Table 4
Grid testing for Test Case-1.

Grid	Radial nodes	Angular nodes	Axial nodes	Total nodes	T_{out} [K]	Difference with exp. ±%
1	55	70	320	12,32,000	447.775	0.30
2	65	80	370	19,24,000	447.720	0.28
3	75	90	420	28,35,000	447.651	0.27

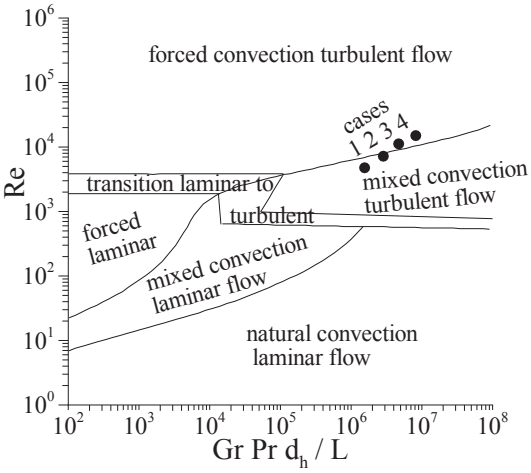


Fig. 4. Regimes of natural, forced and mixed convection for flow through horizontal tubes according to Metais and Eckert [39].

respectively, A_{ap} is the aperture are of the reflector ($A_{ap} = 39$ [m^2]) and I_b is the direct normal irradiance in [W m^{-2}]. According to Dudley et al. [1] the LS-2 collector performance for cermet with vacuum in the annulus may be approximated by the following equation:

$$\eta\% = K[73.3 - 0.007276\Delta T] - 0.496\frac{\Delta T}{I_b} - 0.0691\frac{\Delta T^2}{I_b} \quad (24)$$

and the performance for cermet with air in annulus respectively by:

$$\eta\% = K[73.4 - 0.00803\Delta T] - 9.68\frac{\Delta T}{I_b} - 0.0669\frac{\Delta T^2}{I_b} \quad (25)$$

where K is the incident angle modifier, here taken $K = 1$ [21] and ΔT is the temperature difference above ambient air temperature, $\Delta T = \frac{1}{2}(T_{in} + T_{out}) - T_{amb}$.

3. Validation

In the absence of an experimental study using nanofluids in a LS-2 module, two experimental cases that utilize these two characteristics individually were selected to validate the current numerical study.

3.1. PTC case (LS-2 module)

The experimental data from Dudley et al. [1] for an LS-2 module with Cermet selective coating and vacuumed annulus were selected to validate the present numerical configuration in ANSYS – CFX. These test cases (Syltherm 800 as the working fluid) will be the basis for the nanofluid simulations that will be presented in Section 4.

Grid independency. Considering test case 1 from Table 4, three different computational grids were studied. The % difference in outlet temperature from the experimental data is 0.30, 0.28, 0.27 respectively. Thus, the resolution of grid 2 can be considered as

sufficient and for the rest of the study all the simulations will be conducted with the aforementioned grid.

Results. As seen in Fig. 5 the computational results are in good agreement with the experimental data of Dudley et al. [1] presented in Table 2. A quantitative comparison is also shown in Table 5. Particularly the relative difference % between the experimental values of T_{out} and the numerical ones are all of them below 0.3%. The corresponding relative difference % values of the efficiency of the PTC are higher (between 1.17 and 7.32%). Considering the uncertainty in the measurements and the assumptions made in the numerical model (convection losses in outer glass tube, correlations for the HTF properties, adiabatic boundary conditions in inlet and outlet surfaces, etc.) the agreement may be characterized as fairly good and sufficient in order to proceed to further simulations using nanofluid as the HTF.

3.2. Nanofluid flow through a circular tube

The numerical study of Fard et al. [43] has been chosen in order to validate the accuracy of the two-phase model for the current simulation model concerning nanofluids. Fard et al. [43] used the experimental setup of Heris et al. [44] and Heris et al. [45] for laminar convective heat transfer of nanofluids in a circular tube under constant wall temperature conditions. In their study both single-phase and two-phase models were implemented to predict temperature and flow fields, as well as to calculate the heat transfer coefficient for a range of Peclet (Pe_{nf}) numbers. Based on the results of their simulations it was concluded that the two-phase approach gives better agreement with experimental measurements for heat transfer rate compared to the single-phase model.

For validation purposes, the above simulations (for $\phi = 2\%$) have been reproduced for Al_2O_3/water nanofluid. In Fig. 6, a comparison of heat transfer coefficient is shown between the experimental data of Heris et al. [44,45], the two-phase model results of Fard et al. [43] (results of single-phase model were not available for the particular combination of Al_2O_3/water nanofluid in their study) and the single-phase and two-phase results of the present numerical configuration. It is obvious that Fard et al. [43] statements about the accuracy of two-phase models are valid.

One of the main advantages of two-phase over single-phase models is the need for fewer empirical correlations for density, viscosity, specific heat capacity and thermal conductivity. For those reasons the two-phase model will be used for all the simulations of the present study.

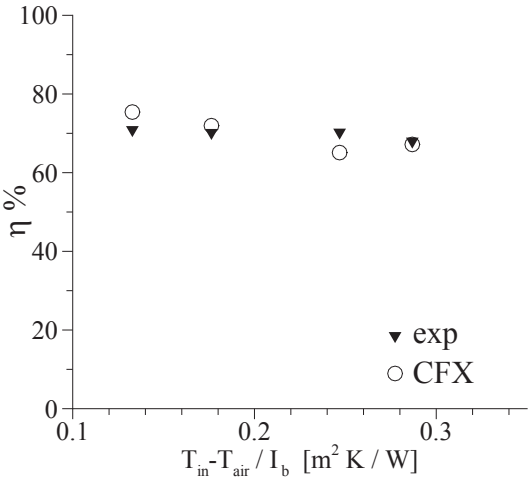


Fig. 5. Efficiency results of the validation case for the LS-2 module with Syltherm 800, compared to the experimental data of [1].

4. Results and discussion

4.1. Flow and temperature results

4.1.1. 3-D results

In this section three dimensional temperature and flow results are presented for typical operating conditions of the LS-2 module, such as the conditions of Case-1, in addition to the effect of the nanoparticles concentration. Fig. 7 in particular, shows the temperature distribution on the outer surface of the solid plug in addition to the streamtraces for the entire receiver length.

Fig. 7a shows the absorbing surface temperature results without the addition of nanoparticles. Near the entrance of the receiver, it seems that there is a uniform circumferential temperature distribution, which develops to non-uniform closer to the outlet. Due to buoyancy effects, hotter regions are evident at the upper side of the solid plug, indicative of loss of axial symmetry. Fig. 7b and c, which correspond to $\phi = 2\%$ and $\phi = 4\%$ respectively, show the same qualitative results for temperature with the only exception being, the higher temperature values as ϕ increases.

The streamtraces presented in Fig. 7, are evidence of the mixed convection regime resulting from the operating conditions. More specifically, even though the fluid seems to move parallel to the X-axis in the entrance region, further downstream its streamtrace starts to bend indicating a tortuous flow path in the annulus. The anti-clockwise rotation of the nanofluid is more intense as ϕ values are increasing. For $\phi = 0\%$ (Fig. 7a), the fluid completes half revolution along the entire length, whereas, for $\phi = 4\%$ (Fig. 7c) it completes a full revolution around the solid plug indicating an increase of buoyancy effects.

4.1.2. 2-D results

Vector profiles. Since the above analysis indicates non-axisymmetric temperature and velocity profiles, it is more illustrative to present similar results in cross-sectional slices. Fig. 8 shows the vector distribution on such a slice, at the middle of the X-axis for $\phi = 0\%$ and for Case 1. The vectors show only the secondary flow in the circumferential direction, since only the u_y and u_z velocity components were taken into account. The first conclusion, is the anti-clockwise rotation along the entire perimeter of the HTF domain as already stated in the previous section. No secondary eddies are present but the radial velocity profile varies along with the circumferential angle. Particularly, at the right side of the HTF domain ($-90° < a < 90°$, see Fig. 2), which is the side with the lowest solar flux values, the velocity profiles closely resemble a laminar parabolic profile. On the contrary, at the left side of the HTF domain ($90° < a < 270°$), there is an acceleration closer to the outer wall. The above observation is expected, since the outer right solid region absorbs nearly the entire amount of solar heat (see Fig. 3b).

The above qualitative vector characteristics do not change across the receiver length, not even with the presence of nanoparticles. For that reason, the authors considered it as not necessary to present additional slices with vector plots from other cases or operating conditions.

Temperature contours. In order to examine the spatial variation of temperature across the receiver length and the effect of nanoparticle concentration, Fig. 9 presents the temperature contours on Y-Z planes at three different positions along the X-axis ($X/L = 0.2$, 0.5, 0.8), for $\phi = 0$, 2, 4% of Case 1. As expected, due to wall heat flux, the nanofluid temperature values rise as it travels across the X-axis. Nevertheless, the spatial temperature distribution over each Y-Z plane is not uniform. Near the entrance (Fig. 9 for $X/L = 0.2$) the maximum temperature values develop mainly on the left outer side

Table 5
Comparison of experimental data from Dudley et al. [1] *and simulation results.*

Case	I_b [W m^{-2}]	$T_{in} - T_{air}/I_b$ [Km^2/W]	$T_{out,exp}$ [K]	$T_{out,CFX}$ [K]	T_{out} *diff* %	η_{exp} %	η_{CFX} %	η *diff* %
1	968.2	0.133	446.45	447.72	0.28	70.90	75.40	6.35
2	982.3	0.176	492.65	493.14	0.10	70.17	71.94	2.52
3	909.5	0.247	542.55	541.21	0.25	70.25	65.10	7.32
4	937.9	0.287	590.05	589.75	0.05	67.98	67.18	1.17

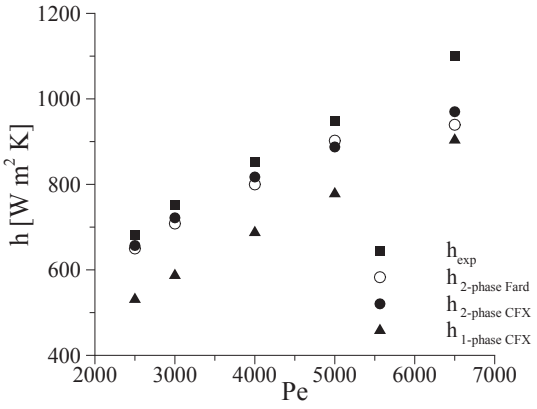


Fig. 6. Comparison of heat transfer coefficients for a range of different Peclet numbers for Al_2O_3/water nanofluid as the heat transfer fluid ($\phi = 2\%$).

of the annulus. The aforementioned behavior is justified, since on that side greater values of heat flux are imposed. As the nanofluid continues its movement through the annulus (Fig. 9 for $X/L = 0.5$, 0.8) buoyancy effects increasingly influence the temperature distribution, shifting the locus of temperature maximum towards the top of the Y-Z slices. The nanoparticle concentration does not seem to affect the position of temperature maximum but leads to an overall rise in the temperature values.

Heatlines visualization. Kimura and Bejan [46] proposed a new approach to visualizing the transfer of heat by fluid flow. Their approach represents the energy-analog of the use of "stream function" and "streamlines" to visualize fluid flow, whereby the so called *heatlines* are employed. In a two dimensional convection heat transfer configuration, the transport of energy through the flow field is a combination of both thermal diffusion and enthalpy flow. For any such field, one can define a function $H(y,z)$ such that the net flow of energy (thermal diffusion plus enthalpy flow) across each $H = constant$ line is zero. The heat function $H(y,z)$ is defined as:

$$\frac{\partial H}{\partial z} = \rho C_p u_y T - k\frac{\partial T}{\partial y}, \quad \text{net energy flow in the Y} - \text{direction} \tag{26}$$

$$-\frac{\partial H}{\partial y} = \rho C_p u_z T - k\frac{\partial T}{\partial z}, \quad \text{net energy flow in the Z} - \text{direction} \tag{27}$$

According to the heat function definition any $H = constant$ heatline is locally parallel to the direction of net energy flow through the convection field.

In Fig. 10 the heatlines on a Y-Z plane midway along the X-axis, for $\phi = 0\%$ of Case 1 are presented, whereas in Fig. 11 those on a X-Y plane at $a = 0°$ and 180°, for $\phi = 0\%$ and Case 1 are shown. Heatlines in both figures reveal that the heat is transferred mainly by convection, since they are very similar to the fluid motion streamlines. It seems that the temperature gradient terms in Eqs. (26) and (27), are relatively small even for the secondary flow movement of

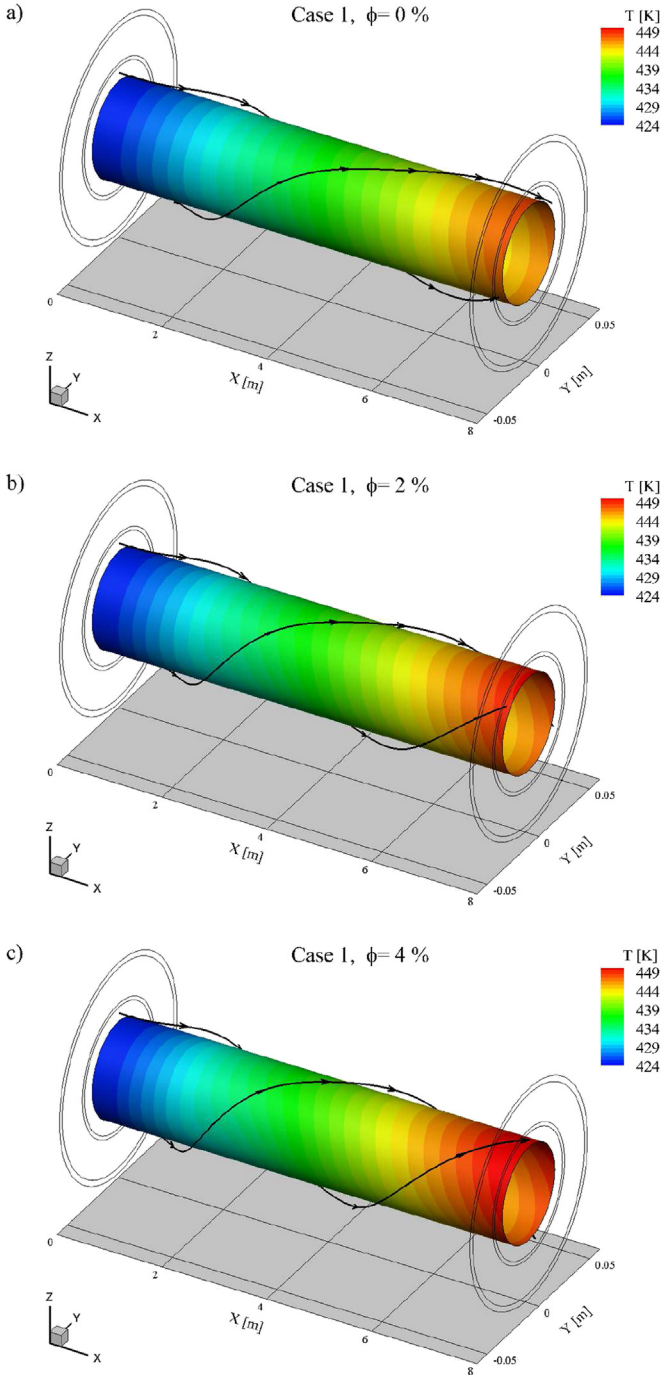


Fig. 7. Streamlines and temperature contours on the surface of the solid plug, for $\phi = 0\%$ (a), $\phi = 2\%$ (b) and $\phi = 4\%$ (c), of Case 1.

Fig. 10. This fact is in accordance with Fig. 4, suggesting that the parameters of the current study lie at the border between forced and mixed convection.

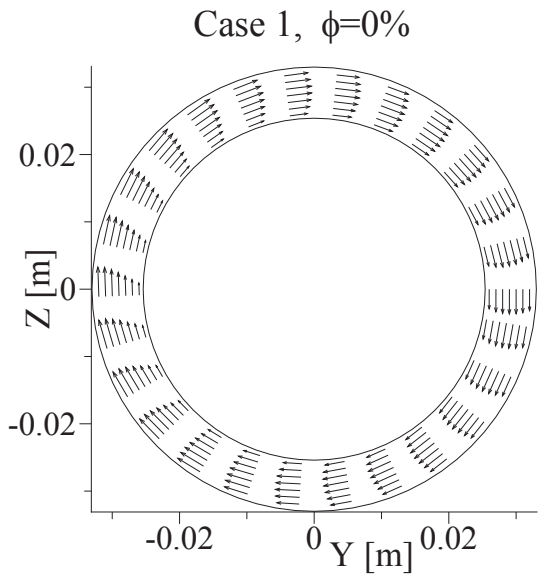


Fig. 8. Vectors on a Y-Z plane midway along the X-axis ($X = 3.9$[m]), for $\phi = 0\%$ of Case 1.

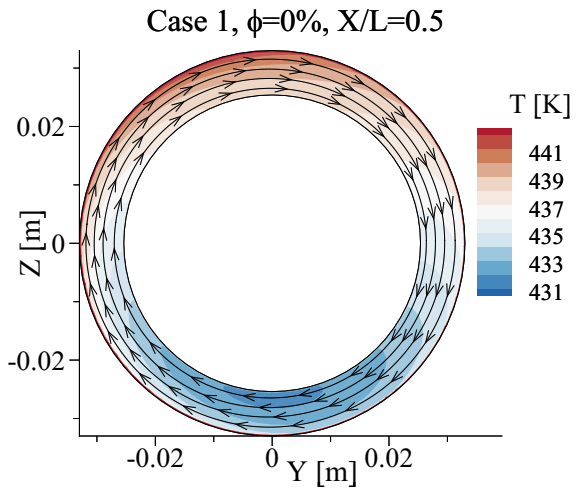


Fig. 10. Heatlines on a Y-Z plane midway along the X-axis, for $\phi = 0\%$ and Case 1.

that the addition of nanoparticles in the HTF causes the temperature to rise for all angles. The maximum temperature difference

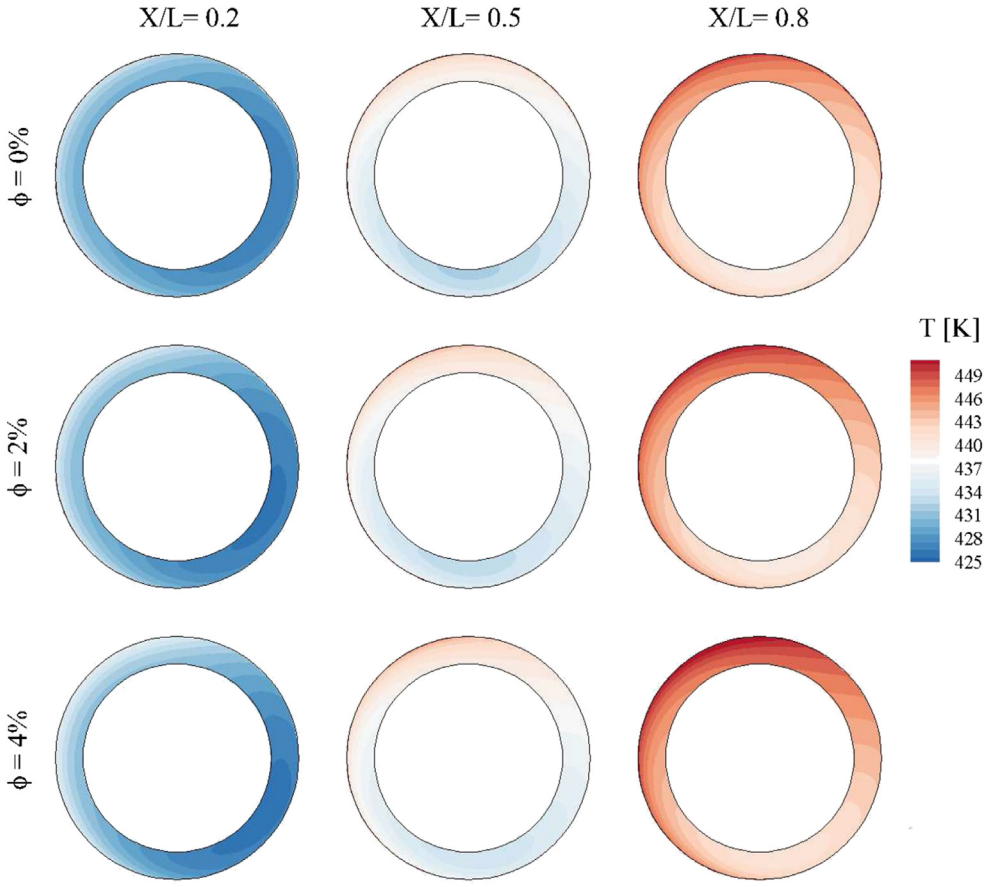


Fig. 9. Temperature contours on a Y-Z plane at selected locations along the X-axis, for $\phi = 0\%$ of Case 1.

4.1.3. 1-D results

In order to evaluate the circumferential variations of temperature along the receiver length, Fig. 12 presents the axial variation of temperature (T_{HTF}) at the center ($r^* = 0.5$) of the annular area, at four different angles, for $\phi = 0\%$ and $\phi = 4\%$ of Case 1. It is evident occurred at 90° angle (2.75 [K]), whereas the minimum temperature difference at 270° angle (2.15 [K]). As seen in Fig. 12a, a non-uniform circumferential temperature distribution develops already at the inlet of the HTF (in ascending temperature values: 0°, 270°, 90°, 180°). At $X = 2.3$ [m], the temperature values at 0° and

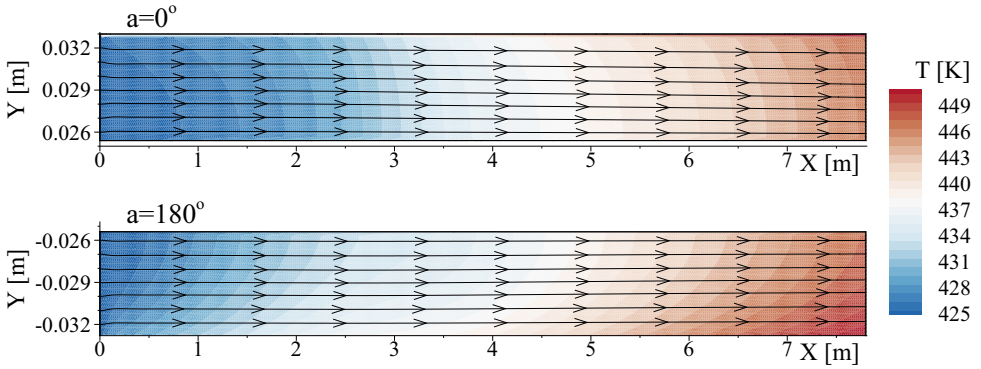


Fig. 11. Heatlines on a X-Y plane at $a = 0°$ and $a = 180°$, for $\phi = 0\%$ of Case 1.

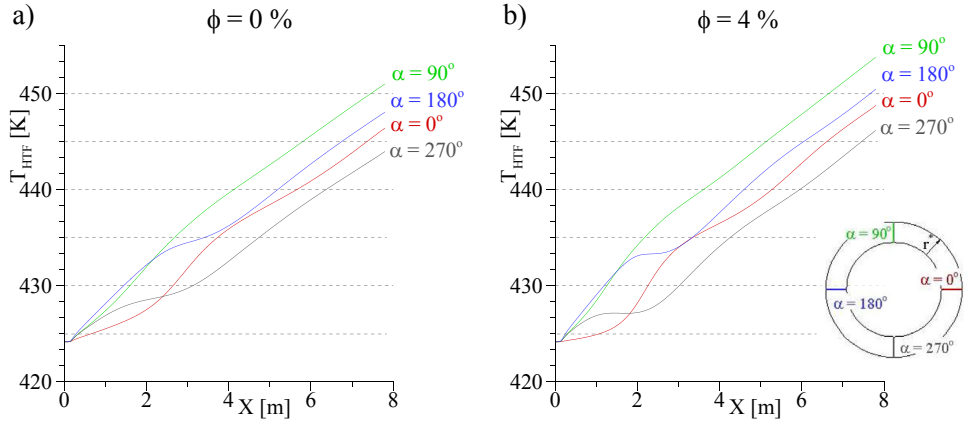


Fig. 12. Temperature profiles along X-axis at four different angular positions, for $\phi = 0\%$ and $\phi = 4\%$ of Case 1.

90° exceed those at 270° and 180° respectively and this trend persists for the remaining receiver length. For the case with nanoparticles of $\phi = 4\%$ concentration (Fig. 12b) this transition occurs earlier, namely at $x = 1.8$ [m], and this is an outcome of the enhanced heat transfer due to the Al_2O_3 nanoparticles. Despite the presence of nanoparticles, at the end of the receiver tube, the position of temperature maxima and minima are unaffected. This is probably due to fact that the non-uniformity of solar flux overshadows the nanoparticle effect on heat transfer.

4.2. Parametric evaluation of nanoparticles

As already mentioned in Section 2, numerical simulations were conducted with a two-phase model for all test cases and for five different nanoparticle concentrations ($\phi\% = 0, 0.5, 1, 2, 4$). A total number of 20 simulations cases thus resulted, enabling a broad parametric study of nanoparticle concentration. Fig. 13 shows the collector overall efficiency ($\eta\%$) against nanoparticle concentration ($\phi\%$) for all test cases. The corresponding lines are the linear approximation. In Fig. 13 the variable $(T_{in} - T_{air})/I_b$ is used in the abscissa, as this is a very common normalized variable used in the literature to plot collector efficiency results [47]. It is apparent that a strong correlation exists between $\phi\%$ and $\eta\%$, namely $\eta\%$ increases linearly with the increase of $\phi\%$. In particular, the values of the a_1 coefficient of the linear approximation equation ($\eta = a_1\phi + a_0$) are 153.4, 162.4, 169.1, 179.2 for $(T_{in} - T_{air})/I_b = 0.13, 0.17, 0.24, 0.29$ respectively. Higher values of this variable are subject to a more intense increase of $\eta\%$ as $\phi\%$ values increases.

The above results, could be presented from a different

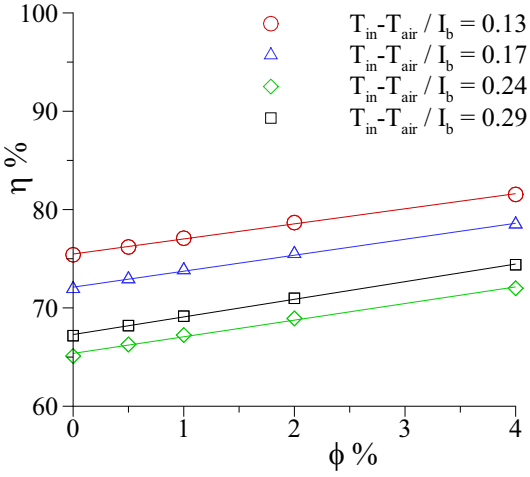


Fig. 13. Collector efficiency ($\eta\%$) against nanoparticle concentration ($\phi\%$) for various inlet temperatures. The solid lines correspond to linear approximations.

viewpoint, as the percentage increase in efficiency $(\eta - \eta_{0\%})/\eta_{0\%}$ due to the addition of nanoparticles against inlet temperature for various particle volume concentrations (Fig. 14). According to the aforementioned figure, $\phi = 0.5\%$ could result in an average increase of 1.4%. As $\phi\%$ increases so does the efficiency, reaching its maximum value of 9.6% (on average) for $\phi = 4\%$. Even though previous studies with Al_2O_3/water nanofluids have shown higher efficiency improvements in flat plate collectors, the present results

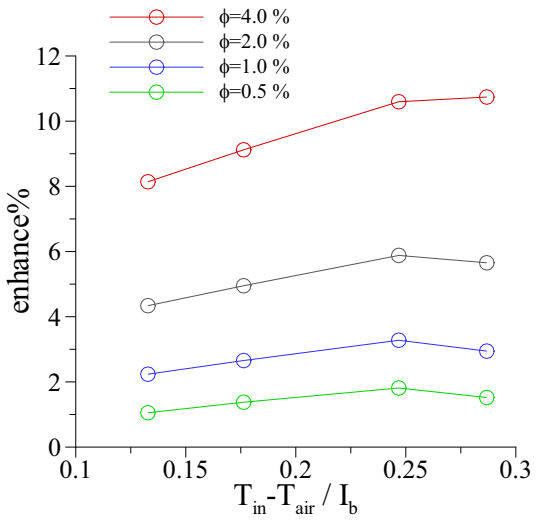


Fig. 14. Percentage increase in efficiency due to the addition of nanoparticles against inlet temperature for various particle volume concentrations.

are more in accordance with those reported in the literature for synthetic oils as the base fluid [13,14].

5. Conclusions

In this study the numerical results of a Parabolic Trough Collector system with Syltherm 800/Al_2O_3 nanofluid as the heat transfer fluid were presented. It was concluded that the two-phase approach has an advantage in terms of accuracy, in contrast to the single-phase approach which is more common in similar studies. A total number of 20 different simulation cases were performed for a wide range of operating conditions and nanoparticle (Al_2O_3) concentrations. As expected, the presence of nanoparticles enhanced heat transfer and increased the collector efficiency. This increase was associated with detailed temperature and velocity fields, showing enhanced mixed convection effects for higher nanoparticle concentrations. According to the parametric evaluation on the LS2 module efficiency, even a 10% boost in efficiency is possible for Al_2O_3 concentration of 4%. The result is in agreement with previous relevant studies which found enhancements of this order of magnitude by using nanofluids in parabolic trough collectors.

Acknowledgment

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Appendix A. Material Properties

Table Appendix A.1
Properties of Pyrex glass

Property	Value
Transmittance	0.95
Refractive index	1.474
Density [g cm^{-3}]	2.23
Specific heat [J kg^{-1}] K^{-1}	750
Thermal conductivity [W m^{-1} K^{-1}]	1.14

Table Appendix A.2
Correlations for the Syltherm 800 properties according to Dow Chemical Company [26].

Property	C_p [J kg^{-1} K^{-1}]	ρ_{in} [kg m^{-3}]	k [W m^{-1} K^{-1}]	μ [Pa s]
a_0	1.10787577E3	1.26903060E3	0.19011994	8.486612E-2
a_1	1.70742274	−1.52080898	−1.88022387E-4	−5.541277E-4
a_2	0	1.79056397E-3	0	1.388285E-6
a_3	0	−1.67087252E-6	0	−1.566003E-9
a_4	0	0	0	6.672331E-13

Table Appendix A.3
Al_2O_3 nanoparticles properties

Property	Value
ρ_{in} [kg m^{-3}]	3850
C_p [J kg^{-1} K^{-1}]	$1046 + 0.174\,T - 2.79 \times 10^7\,T^2$
k [W m^{-1} K^{-1}]	$5.5 + 34.5e^{-0.0033(T-273)}$
μ [Pa s]	1.0E-08

Nomenclature

a	thermal diffusivity [m^2 s^{-1}]
C_p	specific heat under constant pressure [kg^{-1} K^{-1}]
d_h	hydraulic diameter $d_h = 2(r_{out} - r_{in})$
g	magnitude of the gravitational acceleration [m s^{-2}]
Gr	Grashof number (dimensionless), $Gr = g\beta d_h^4 q''/\kappa\nu^2$
I_b	direct normal irradiance [W m^{-2}
k	thermal conductivity [W m^{-1} K^{-1}]
L	length of the LS2 module [m]
p	pressure [N m^{-2}]
Pr	Prandtl number (dimensionless), $Pr = \nu/\alpha$
$\dot{Q}_{in}$	volumetric flow rate, $\dot{Q}_{in} = u_{in}h_{in}A_{in}$ [m^{3}s^{-1}]
q''	heat flux on absorber tube [W m^{-2}]
r	radius [m]
Ra	Rayleigh number (dimensionless), $Ra = GrPr$
Re	Reynolds number (dimensionless), $Re = u_{in}d_h/\nu = \dot{m}_{in}d_h/\nu\rho A_{in}$
Ri	Richardson number (dimensionless), $Ri = Gr/Re^2$
t	physical time [s]
T	local temperature [K]
X,Y,Z	coordinate distance along the length, height and width of the tank [m]
u,v,w	Cartesian velocity components [m s^{-1}]
u_{in}	inlet velocity [m s^{-1}]

Greek Symbols

α	Angle
β	coefficient of thermal expansion $\beta = -\frac{1}{\rho}\frac{\partial\rho}{\partial T}$, [K^{-1}]
η	Efficiency
ν	kinematic viscosity[m^2 s^{-1}]
ρ	density [kg m^{-3}]
Ω	volume [m^3]
ϕ	volume fraction of the nanoparticles

Subscripts

amb	Ambient
f	Fluid
in	Inlet
nf	Nanofluid
s	Solid
out	Outlet

Acronyms

CFD	Computational Fluid Dynamics
FEM	Finite Element Method
HTF	Heat Transfer Fluid
LCR	Local Concentration Ratio
MCRT	Monte Carlo Ray Tracing
MWCNT	Multi Walled Carbon Nano Tubes
PTC	Parabolic Trough Collector

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